

PegasusX — Capabilities: Math, Logic, Algorithms (CODE)

SOURCE OF TRUTH: Go packages under `apps/backend-go` (not other markdown).

Inventory of *which* APIs exist: [FEATURES_BY_APP_ROLE.md](#)

Order transitions: [ORDER_FLOW_AND_EDGE_CASES.md](#)

Each block cites the implementing file/function. Money is `int64` minor units in structs (`UnitPrice / TotalMinor / *Minor` fields).

0. Shared engines

0.1 Credit available — `credit.Profile.Available`

File: `apps/backend-go/credit/types.go`

```
Available = max(0, CreditLimitMinor - CurrentBalanceMinor - ReservedMinor)
```

Statuses (`credit.Status`): `INACTIVE`, `ACTIVE`, `FROZEN`, `CLOSED`, `BLACKLISTED`.

RiskTier (`LOW|MEDIUM|HIGH|BLOCK`) exist as types; leave-on-credit **does not use them** (see §0.2).

What / solution: Headroom for credit orders and credit-leave without over-extension.

Edges: Negative headroom clamped to 0; reserved holds block double-spend across open orders.

0.2 Credit leave gate — `order.CanLeaveOnCredit`

File: `apps/backend-go/order/credit_guard.go`

Logic:

1. `profile != nil` and `profile.Status == ACTIVE`
2. `profile.Available() >= order.TotalMinor`
3. If `cfg.MaxAutoCreditMinor > 0` and `order.TotalMinor > MaxAutoCreditMinor` → reject

Comment in code: *"Credit risk scoring / RiskTier is intentionally not used (Phase A removal)."*

Call sites `hardcode MaxAutoCreditMinor: 50000000` in `retailer_shop_closed.go` and `worker_shop_closed.go`.

0.3 Inventory reserve — `order.ReserveLineItemsInTxn`

File: `apps/backend-go/order/inventory_reservation.go`

Logic:

1. Aggregate qty by SKU (duplicate lines summed)
2. Read `SupplierInventoryV2` (`QuantityOnHand`, `QuantityReserved`)
3. Require `QuantityOnHand - QuantityReserved >= quantity`
4. Write `QuantityReserved += quantity`
5. Missing SKU → `ErrInventoryExhausted`
6. Marker row `OrderStockReservationMarkers`

Backfill: `BackfillScheduledReservations` reserves for `MANUAL_PREORDER` in `SCHEDULED|AUTO_ACCEPTED` missing markers.

0.4 Inventory release / stale

Files: `inventory_release.go`, `inventory_stale_release.go`

Release: `QuantityReserved = max(0, QuantityReserved - qty)`.

Stale job: select `Status = PENDING` with old `UpdatedAt`, cancel via cancel path (not direct decrement) — avoids double-release with `AWAITING_PAYMENT`.

0.5 Dispatch volume & capacity

Files: `dispatch/constants.go`, `dispatch/volume.go`, `dispatch/binpack.go`, `dispatch/capacity_recommend.go`

```
TetrisBuffer = 0.95
DefaultTruckVolumeVU = 150.0
defaultUnitVolumeVU = 1.0
effectiveCap = MaxVolumeVU * TetrisBuffer
```

Order volume: sum over lines `max(qty,1) * unitVU` where `unitVU` from line → product catalog → default 1.0 (`VolumeSourceCatalog` vs `default_1_0`).

SelectBestVehicle / binpack reject when order volume exceeds `MaxVolumeVU * TetrisBuffer`. Capacity recommend unselects until under effective cap.

0.6 Approach geofence

File: `apps/backend-go/proximity/geofence.go`

```
DeliveryApproachRadiusM = 500.0
```

Wired as `deliveryGeofenceMeters` in `order/service.go`. Settlement proximity comments in service reference **100 m / H3** unlock path (`ProximityUnlockedAt` / `proximity-unlock` route).

0.7 Shop-closed grace

File: `order/service.go` `NewService`

If `ShopClosedGrace` <= 0 → default `5 * time.Minute`.

Bootstrap sets via `shopClosedGraceDuration()` (`bootstrap/bootstrap.go`).

Worker `worker_shop_closed.go` polls `SHOP_CLOSED_PENDING` where `ShopClosedGraceEndsAt` <= now (limit batch), then `DecideShopClosedTimeout` with `MaxAutoCreditMinor`: `50000000`.

0.8 Fiscal constants — `order/fiscal.go`

```
FiscalOFDTimeout = 8 * time.Second
```

Cash variance (`emitCashVariance`):

```
shortfall = max(0, expected - received)
overage   = max(0, received - expected)
```

Emit `CASH_SHORTFALL` or `CASH_OVERAGE` only if either > 0.

Force-complete: roles `ADMIN` / `WAREHOUSE_ADMIN` (route in `orderroutes`); statuses handled in `ForceCompleteOrder` (`FISCAL_*` / reconciliation paths in `fiscal.go`).

0.9 Partial offload — `order/partial_offload.go`

Statuses: `FULL` | `PARTIAL` | `NONE` | `RETURNED`.

Reasons: `DAMAGED` | `MISSING` | `SHOP_REFUSED` | `CAPACITY` | `OTHER`.

Invariant in `ApplyPartialOffloadLines`:

```
DeliveredQty + RemainingQty == Quantity (per touched line)
delivered_minor = Σ UnitPrice × DeliveredQty
remaining_minor = Σ UnitPrice × RemainingQty
```

0.10 Claim window — `claims/service.go` + `order/claim_window.go`

```
DefaultPostDeliveryClaimWindow = 48 * time.Hour
```

Env override `CLAIM_WINDOW_HOURS` ; supplier/warehouse policy fields snapshotted onto order at COMPLETED (`ClaimWindowHours` , `ClaimWindowEndsAt` , `ClaimWindowPolicySource`).

0.11 Claim pricing — `claims/pricing.go`

1. `AggregateClaimLines` merges duplicate SKUs (overflow-safe add).
2. Order SKU index: weighted avg unit price
`unit = valueSum/qty` then half-up if `rem*2 >= qty` (equiv. `floor((valueSum+qty/2)/qty)`).
3. `ClaimedQtyBySKU` counts prior claims in `OPEN|UNDER_REVIEW|APPROVED|RESOLVED` (not `REJECTED`).
4. Require claim `qty ≤ order qty – claimed`; amount = $\sum \text{qty} \times \text{unit}$ (overflow-checked `mulInt64 / addInt64`).

0.12 Claim eligibility — `claims/eligibility.go`

Requires order status `COMPLETED` (constant `OrderStatusCompleted = "COMPLETED"`). Photo-required types enforced in eligibility logic (`DAMAGED` family — see function body). Window from snapshot or default 48h.

0.13 Idempotency — `order/idempotency_guard.go`

SHA-256 body hash + Idempotency-Key headers; mismatch → conflict; used by `HandleCollectCash` among others.

0.14 State machine — `order/state_machine.go`

Canonical transitions only (full table in `ORDER_FLOW` doc). Same-status allowed. No soft `ARRIVED→COMPLETED` (comment `ADR-009`).

1. Retailer capabilities (logic tied to routes)

Capability	What it does	Solution	Logic / math	Edges	Evidence
Checkout + reserve	Create order, hold stock	Prevent oversell	\$0.3 in same txn as create	OOS → <code>inventory_exhausted</code> ; empty warehouseID no-op reserve	<code>inventory_reservation.go</code> , <code>paymentroutes</code> checkout
Credit checkout / leave	Use trade credit	Buy without doorstep cash	\$0.1–0.2	Frozen/blacklisted; max auto 50_000_000 minor	<code>credit/*</code> , <code>credit_guard.go</code>
Shop-closed respond	Resolve closed shop	Unstick delivery	Routes → status via <code>retailer_shop_closed</code> handlers	Grace 5m default; credit leave needs <code>ACTIVE+Available</code>	<code>retailer_shop_closed.go</code> , <code>worker_shop_closed.go</code>
Preorder confirm/edit/reject	Future-dated commitment	Plan ahead of T-1	<code>ConfirmationStatus</code> + <code>SCHEDULED/AUTO_ACCEPTED</code> → <code>PENDING</code> promote	Edit/cancel locks in <code>preorder_policy/sweeper</code>	<code>preorder_*.go</code>
Claims	Post-delivery redress	Fair chargeback amounts	\$0.10–0.12	Not <code>COMPLETED</code> ; window; SKU caps; overflow	<code>claims/*</code>
Cart / quote	Sync cart, price quote	Cross-device cart	<code>Sum qty * unit_price_minor</code>	Stale prices	<code>retailer cart/quote</code> handlers
POS / stock / shifts	Store OS packs	In-store ops	int64 tender amounts; stock movements	Capability pack gates on JWT	<code>retailerroutes</code> <code>stock/pos/shifts</code>
Auto-order / AI	Rules + predictions	Reduce manual reorders	Settings patches + <code>AI_PREORDER</code> source	<code>confirm/reject-ai</code> routes	<code>retailerroutes</code> AI + auto-order

Capability	What it does	Solution	Logic / math	Edges	Evidence
Confirm cash	Retailer confirms COD intent	Align with driver collect	POST /v1/delivery/confirm-cash RoleRetailer	Race with driver collect	orderroutes

2. Supplier (ADMIN) capabilities

Capability	What / solution	Logic / math	Edges	Evidence
Vet orders	Accept/reject before load	Reject → cancel + release inventory	Concurrent vet	supplierroutes vet + inventory_release
Dispatch preview/execute	Build routes under capacity	§0.5 TetrisBuffer / binpack / optional optimizer	Over-cap unselect; freeze locks (warehouse)	dispatch/* , supplier dispatch routes
Payment bypass	Token for doorstep	Driver confirms bypass	Scope to awaiting payment	supplier payment-bypass + driver confirm-payment-bypass
Claims approve/reject	Adjudicate OPEN claims	Pricing from order lines §0.11	Double settle idempotency	orderroutes claims + claims service
Credit profiles/program	Set limits/terms/hold	§0.1 Available	Policy v2 flags in credit package	creditroutes
Cash recon accept/write-off	Close driver cash bags	Ledger entries	Shortfall write-off	cashreconroutes
Inventory import	CSV → apply	Session pipeline + Kafka worker	Bad rows; freeze locks	supplier inventory import + ai-worker
Control tower / playbooks	Scored exceptions	/v1/control-tower/* RoleAdmin	Portal-first UI	controlltowerroutes
Negotiations	Qty negotiate resolve	Routes exist; may return disabled	Product flag	supplier negotiations routes

3. Warehouse capabilities

Capability	What / solution	Logic / math	Edges	Evidence
Dispatch hub	Depot-owned plan/execute	Same §0.5; locks /dispatch-lock*	Role WarehouseAdmin for most ops	warehouseroutes
Order delay/reject/overflow/propose	Reshape commitment	Status DELAYED / cancel / proposal PENDING_WAREHOUSE	Inventory release on reject	warehouse ops order routes + order package
Returns inbound	Physical dock	Scan/confirm sessions	Wrong WH; barcode	returnsroutes
Reverse logistics	Claim-linked receive	Task receive	RoleWarehouse only	creditnoteroutes
Claims adjudicate	Local approve/reject	Same claims service	WarehouseAdmin	orderroutes
Reassign	Move order without payload UI	Capacity check	Concurrent seal	warehouse reassign routes
Return policy CRUD	Policy document	Admin/WarehouseAdmin	No dedicated shell href observed	warehouseroutes return-policy

4. Factory capabilities

Capability	What / solution	Logic / math	Edges	Evidence
Manifest load/seal/dispatch/complete	Outbound correctness	Manifest mutations + outbox events	Concurrent seal	factoryroutes
Rebalance / cancel-transfer	Hot correction	Manifest mutation	After LOADED	factory manifests routes
Supply requests / transfers	Factory↔warehouse stock	Transfer state machine in factory package	FACTORY_DRIVER arrive via driver API	factory + driver supply-transfers
Insights	Demand pull view	Calls warehouse replenishment insights	Cross-package scope	warehouse insights route

5. Payload capabilities

Capability	What / solution	Logic / math	Edges	Evidence
Seal / seal-completed / seal-all	Driver-ready truck	Manifest seal transitions	seal-all registered; verify client use	payloaderroutes
Inject order	Late add	Capacity GET /v1/payload/capacity/{vehicleID}	Over VU	payload capacity + inject
Reassign	Suggest/apply truck move	Capacity + recommend	After seal	recommend/reassign routes
Exceptions	Cannot-load etc.	Exception rows + optional delivery exception-report	Dual paths	payload manifest-exception
Inbound returns	Dock	Shared returns inbound	Role PAYLOAD	returnsroutes

6. Driver capabilities

Capability	What / solution	Logic / math	Edges	Evidence
Arrive	Mark ARRIVED	Geofence §0.6	geofence_violation	POST /v1/delivery/arrive
Proximity unlock	Unlock settlement	100 m / H3 / unlock token (service comments)	proximity_required	proximity-unlock + service
QR / offload / deliver	→ AWAITING_PAYMENT	ValidateStatusTransition	Replay QR	scan-qr, deliver, confirm-offload
Collect cash / complete	→ FISCALIZING	§0.8 variance; ADR-009	Idempotency §0.13; short cash	collect-cash, complete, fiscal.go
Credit leave	→ DELIVERED_ON_CREDIT	§0.2	Cannot COMPLETED until FISCALIZING	credit-leave, credit-delivery
Partial offload	Line qty adjust	§0.9	Invalid status after fiscalizing	partial_offload.go
Shop closed	→ SHOP_CLOSED_PENDING	Grace §0.7	Timeout worker matrix	shop-closed handlers + worker
Bypass offload	Supervised → AWAITING_PAYMENT	Token + photo opts in TransitionOpts	Token mismatch	bypass-offload
Fiscal retry	FISCAL_FAILED → FISCALIZING	OFD 8s timeout	Max attempts in fiscal package	fiscal/retry
Offline sync	Batch upload	/v1/sync/batch	Clock skew checks on cash	sync + idempotency
Supply transfer arrive	Factory leg	Driver/FACTORY_DRIVER	Wrong transfer id	supply-transfers
Force-complete	N/A for driver	—	RoleAdmin/WarehouseAdmin only	orderroutes

7. Outbox / concurrency (platform)

Mutations that change order status typically: Spanner RW txn + outbox emit (`emitOrderStatusChanged` in `order/service.go`) + version optimistic concurrency. Illegal transitions return `ErrInvalidStatusTransition` .

8. Rebuild rule

When updating this file:

1. Open the cited `.go` file.
2. Copy formulas/constants from source.
3. Do not import wording from other `.md` files.

Known code nuances:

- `RiskTier` types remain; `CanLeaveOnCredit` ignores them.
- Claim prior-qty statuses $\neq$ residual claim money filters (see `claims/residual.go` if present).
- Shop-closed wire status string is `SHOP_CLOSED_PENDING` .